

|    |   |                                                                                                                                                                                                                                                                                                                                 |
|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11 | C | $F_n = ma$ $mg = ma$ <p>Since <math>g = -\frac{d\phi}{dr} = -\frac{\Delta\phi}{\Delta r}</math> for a uniform gravitational field</p> $\frac{m\Delta\phi}{\Delta r} = \frac{E}{x} = ma$ $a = \frac{E}{mx}$ <p>The direction of <math>F_n</math> and <math>g</math> is from higher potential to lower potential, i.e. P to Q</p> |
| 12 | D | <p>For simple harmonic motion, <math>a = -\omega^2 x</math>. Hence the gradient of an <math>a</math> vs <math>x</math> graph is <math>-\omega^2</math></p>                                                                                                                                                                      |